

Customer Churn & Retention Analysis

IBM Telco Customer Churn Dataset | R | Customer Analytics

1. Project objective

Identify patterns associated with customer churn and translate the findings into practical retention insights for a telecommunications business.

2. Dataset

The project uses the IBM Telco Customer Churn dataset with 7,043 customer records. The analysis focuses on contract type, payment method, tenure and monthly charges.

3. Analytical workflow

The R workflow imports the customer-level CSV, converts churn, contract and payment method to factors, calculates churn rates by contract and payment method, compares tenure and monthly charges by churn status, and applies Chi-square tests to assess whether churn is associated with contract type or payment method.

4. Key analytical questions

Question	Method
Does churn vary by contract type?	Churn-rate comparison + Chi-square
Does churn vary by payment method?	Churn-rate comparison + Chi-square
Do churned customers differ in tenure?	Group descriptive comparison
Do churned customers differ in monthly charges?	Group descriptive comparison

5. Interpretation framework

Contract and payment-method differences are useful for retention because they describe customer commitment and billing behaviour. Tenure provides a lifecycle perspective, while monthly charges help identify the commercial value of customers who leave. The supporting charts in the repository visualise these relationships.

6. Business implications

Retention activity should be targeted rather than applied uniformly. Customer segments showing higher observed churn can be prioritised for onboarding support, contract incentives, payment-method education or proactive service outreach. Observational associations should not be treated as causal effects without further testing.

7. Reproducibility

The repository includes analysis.R and supporting PNG visualisations. The original dataset is expected locally and is not redistributed by the project. Running the script with the CSV in the project directory reproduces the descriptive summaries and statistical tests.